

NAWAL SHAHID

Software Engineer

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PROFILE

Innovative and results-driven Software Engineer with a strong foundation in **full-stack development, mobile applications, and data-driven solutions**. Experienced in designing and delivering **robust, user-centric applications** across web and mobile platforms, leveraging technologies like **Python, Java, JavaScript, React, Node.js, and Android Studio**. Skilled in **machine learning, data structures, and software architecture**, with a proven ability to translate complex problems into scalable solutions. Passionate about creating **impactful digital experiences** and continuously exploring emerging technologies to drive innovation.

EDUCATION

Jinnah University for Women, 2022 – 2026

BSc Software Engineering | Jinnah University for Women

- Creative Art Manager - Managed event branding, design, and creative direction for university functions.
- Technical Management Co-Lead - Google Developer Student Club (GDSC)- Led developer events. Mentored peers on real-world projects.

GPA: 3.4/4.0

WORK EXPERIENCE

AJD Marketing (Qatar), Web Developer Apprentice 2023 – 2024

- Developed and maintained responsive websites using React, PHP, SQL, and Tailwind CSS.
- Handled full-stack development, including both frontend UI components and backend logic.
- Applied SEO best practices that boosted site traffic and engagement.
- Debugged and optimized legacy code for performance, accessibility, and responsiveness.
- Produced clean, reusable code and documented solutions for long-term maintainability.

Kazim Trust, Web & Mobile App Developer - DigiLex 2024 – 2025

- Leading the end-to-end development of DigiLex, an ed-tech platform designed to support dyslexic learners through accessible digital solutions.
- Built a full-featured, accessibility-first website using React.js, Node.js, MongoDB, and Tailwind CSS.
- Currently developing the DigiLex cross-platform mobile app using React Native and Firebase, ensuring consistent performance across Android and iOS.
- Collaborated with educators, therapists, and accessibility specialists at Kazim Trust to align digital features with real-world needs and inclusive design standards (WCAG).

PROJECTS

News Classifier — Advanced News Article Classification System,

Python · scikit-learn · Streamlit · TextBlob · BeautifulSoup [🔗](#)

- Built a **production-ready news classifier** for 5 categories (Business, Entertainment, Politics, Sports, Technology) on a **40,000+ BBC article dataset**.
- Achieved **93.1% accuracy** with an ensemble of Naive Bayes, SVM, and Random Forest; added **real-time sentiment analysis** using TextBlob.
- Developed **interactive Streamlit dashboard** with URL input, dataset upload, word clouds, probability histograms, and downloadable CSV.
- Automated **URL extraction & preprocessing**, and implemented **per-prediction explainability** (feature importance, class probability).

Movie Discovery — Android App (Android Studio & Kotlin),

Kotlin · MVVM · Room · Retrofit · Glide · Firebase Auth · Jetpack Navigation [🔗](#)

- Developed a production-grade movie discovery app using **Material 3** and MVVM architecture for maintainable, testable code.
- Integrated **TheMovieDB API** via Retrofit + OkHttp (Gson, logging) to fetch trending, popular, and now-playing movies with pagination and caching.
- Built auth flow with **Firebase Auth (Email + Google Sign-In)** and profile management; supported secure sign-out and token refresh flows.
- Optimized image loading and memory with **Glide**, and improved UX via Jetpack Navigation (safe args) and custom splash activity.

Movie Discovery — Responsive Website,

React · Firebase · REST API · CSS Grid / Flexbox · Responsive Design [🔗](#)

- Built a responsive web frontend to browse, filter (by genre), and search movies; integrated movie details (cast, companies, runtime).
- Implemented Firebase Authentication and a favorites system synced with the mobile app (or local storage if desired).
- Improved UX through lazy loading, client-side caching, and accessible responsive layouts for mobile/tablet/desktop.

TECHNICAL SKILLS

Frontend Development — HTML5, CSS3, Sass, JavaScript (ES6+), TypeScript, React.js, Next.js, Tailwind CSS, Framer Motion, Material UI, Responsive Design, Mobile-First Development, Cross-Browser Compatibility, UI/UX Principles, **Backend Development & Databases** — Node.js, Express.js, RESTful APIs, GraphQL, MongoDB, Firebase Firestore, SQL (PostgreSQL, MySQL), Authentication & Authorization (JWT, OAuth2, Firebase Auth), Cloud Functions, Serverless Architecture, Data Modeling, ORM (Sequelize, Mongoose), **Mobile & Cross-Platform Development** — React Native, Expo, Android Studio, Kotlin, Firebase (Realtime DB, Auth, Cloud Functions), Offline Persistence, Cross-Platform Deployment, App Lifecycle Management, Push Notifications, In-App Authentication, **DevOps & Tools** — Git, GitHub, GitHub Actions, VS Code, Postman, Docker, CI/CD Pipelines, Figma, Netlify, Vercel, Debugging & Logging, Unit Testing (Jest), Integration Testing, Version Control, Agile/Scrum Workflow, **Additional Skills & Competencies** — SEO Optimization, Performance Optimization, Technical Writing, Code Review, Problem Solving, Agile/Scrum Methodologies, Collaboration & Communication, Feature Prioritization, Software Architecture.